

1. A real estate company wants to develop a system that predicts house prices based on square footage, number of bedrooms, and location.

Q: Identify the problem type and outline the step-by-step logic to solve it.

Answer:

Problem Type: Regression

Step-by-step logic:

- **Data Collection:** Collect past data with features such as square footage, number of bedrooms, and location.
- **Data Preprocessing:** Categorical data handling using **label or one-hot encoding**, null values handling.
- **Feature Selection:** Selecting feature and target variable(X,Y).
- **Data Splitting:** Splitting data into training and testing sets using **train_test_split**.
- **Model Selection:** Select regression models such as Multiple linear regression, decision tree, random forest etc...
- **Model Training:** Train the data with different models using **.fit()** method.
- **Model Evaluation:** Evaluate the model using metrics such as **R-squared Score**.
- **Make Predictions:** Use the best model to predict house prices.

2. A bank wants to build a model to detect fraudulent transactions by analyzing customer spending behavior and transaction history.

Q: Identify the problem type and outline the step-by-step logic to solve it.

Answer:

Problem Type: Classification

Step-by-step logic:

- **Data Collection:** Collect past data with features such as spending behavior and transaction history
- **Data Preprocessing:** Categorical data handling using **label or one-hot encoding**, null values handling.
- **Feature Selection:** Selecting feature such as transaction history and target variable fraud or not.
- **Data Splitting:** Splitting data into training and testing sets using **train_test_split**.
- **Model Selection:** Select classification models such as logistic regression, k-nearest neighbour, naive bayes etc...
- **Model Training:** Train the data with different models using fit method
- **Model Evaluation:** Evaluate the model using metrics such as accuracy, f1 score, precision, recall.
- **Make Predictions:** Use the best model to detect transaction is legit or not.

3. A supermarket wants to segment its customers based on their shopping patterns to provide personalized promotions.

Q: Identify the problem type and outline the step-by-step logic to solve it.

Answer:

Problem Type: Clustering

Step-by-step logic:

- **Data Collection:** Collect past data with features such as spending score, income and frequency of purchases.
- **Data Preprocessing:** Data normalization technique and null value handling.
- **Feature Selection:** Selecting feature such as spending score and frequency(X).
- **Model Selection:** Select clustering algorithms such as K-means clustering, affinity propagation, DBSCAN.
- **Determine Optimal Clusters:** Use elbow method and dendrogram to find optimal number of clusters
- **Model Training:** Train the data with different models
- **Analyze Clusters:** interpret and get insights from the clusters
- **Target Marketing:** Use these clusters and make personalized recommendations and promotions

4. A company wants to estimate an employee's salary based on their years of experience, job title, and education level.

Q: Identify the problem type and outline the step-by-step logic to solve it.

Answer:

Problem Type: Regression

Step-by-step logic:

- **Data Collection:** Collect past data with features such as years of experience, job title and educational level
- **Data Preprocessing:** Categorical data handling using **label or one-hot encoding**, null values handling.
- **Feature Selection:** Selecting feature and target variable(X,Y).
- **Data Splitting:** Splitting data into training and testing sets using **train_test_split**.
- **Model Selection:** Select regression models such as Multiple linear regression, decision tree, random forest etc...
- **Model Training:** Train the data with different models using **.fit()** method.
- **Model Evaluation:** Evaluate the model using metrics such as **R-squared Score**.
- **Make Predictions:** Use the best model to predict employees salary.

5. An email provider wants to automatically classify incoming emails as spam or not spam based on their content and sender details.

Q: Identify the problem type and outline the step-by-step logic to solve it.

Answer:

Problem Type: Classification

Step-by-step logic:

- **Data Collection:** Collect past mails with spam and not spam content
- **Data Preprocessing:** Convert text data into numerical format using TF-IDF vectorizer or word embeddings
- **Data Splitting:** Splitting data into training and testing sets using **train_test_split**.
- **Model Selection:** Select classification models such as logistic regression, k-nearest neighbour, naive bayes etc...
- **Model Training:** Train the data with different models using fit method
- **Model Evaluation:** Evaluate the model using metrics such as accuracy, f1 score, precision, recall.
- **Make Predictions:** Use the best model to detect mails as spam or not real time.

6. A business wants to analyze customer reviews of its products and determine whether the sentiment is positive or negative.

Q: Identify the problem type and outline the step-by-step logic to solve it.

Answer:

Problem Type: Classification

Step-by-step logic:

- **Data Collection:** Collect past customer reviews with both positive and neagative.
- **Text processing:** Tokenize the sentences into separate word, remove stopwords and punctuations
- **Data Preprocessing:** Convert text data into numerical format using TF-IDF vectorizer or word embeddings
- **Data Splitting:** Splitting data into training and testing sets using **train_test_split**.
- **Model Selection:** Select classification models such as logistic regression, k-nearest neighbour, naive bayes etc...
- **Model Training:** Train the data with different models using fit method
- **Model Evaluation:** Evaluate the model using metrics such as accuracy, f1 score, precision, recall.
- **Make Predictions:** Use the best model to determine the new reviews are positive or not.

7. An insurance company wants to predict whether a customer is likely to file a claim in the next year based on their driving history and demographics.

Q: Identify the problem type and outline the step-by-step logic to solve it.

Answer:

Problem Type: Classification

Step-by-step logic:

- **Data Collection:** Collect past data with driving history and demographics.
- **Data Preprocessing:** Categorical data handling using **label or one-hot encoding**, null values handling.
- **Feature Selection:** Selecting features and label that is independent and dependent data.
- **Data Splitting:** Splitting data into training and testing sets using **train_test_split**.
- **Model Selection:** Select classification models such as logistic regression, k-nearest neighbour, naive bayes etc...
- **Model Training:** Train the data with different models using fit method
- **Model Evaluation:** Evaluate the model using metrics such as accuracy, f1 score, precision, recall.
- **Make Predictions:** Use the best model to predict customer will claim insurance or not.

8. A streaming platform wants to recommend movies to users by grouping them based on their viewing preferences and watch history.

Q: Identify the problem type and outline the step-by-step logic to solve it.

Answer:

Problem Type: Clustering

Step-by-step logic:

- **Data Collection:** Collect past data with features such as watch history and preferences of genre
- **Data Preprocessing:** Convert categorical data into numerical data,
- **Feature Selection:** Select features such as genre preference and ratings.
- **Model Selection:** Select clustering algorithms such as K-means clustering, affinity propagation, DBSCAN.
- **Determine Optimal Clusters:** Use elbow method and dendrogram to find optimal number of clusters
- **Model Training:** Train the data with different models .
- **Analyze Clusters:** interpret and get insights from the clusters.
- **Target Marketing:** Use these clusters and make personalized movie recommendations.

9. A hospital wants to predict the recovery time of patients after surgery based on their age, medical history, and lifestyle habits.

Q: Identify the problem type and outline the step-by-step logic to solve it.

Answer:

Problem Type: Regression

Step-by-step logic:

- **Data Collection:** Collect past data with features age, medical history and lifestyle habits.
- **Data Preprocessing:** Categorical data handling using **label or one-hot encoding**, null values handling.
- **Feature Selection:** Selecting feature and target variable(X,Y).
- **Data Splitting:** Splitting data into training and testing sets using **train_test_split**.
- **Model Selection:** Select regression models such as Multiple linear regression, decision tree, random forest etc...
- **Model Training:** Train the data with different models using **.fit()** method.
- **Model Evaluation:** Evaluate the model using metrics such as **R-squared Score**.
- **Make Predictions:** Use the best model to predict patients recover time.

10. A university wants to predict a student's final exam score based on study hours, attendance, and past academic performance.

Q: Identify the problem type and outline the step-by-step logic to solve it.

Problem Type: Regression

Step-by-step logic:

- **Data Collection:** Collect past data with features study hours, attendance, and past academic performance.
- **Data Preprocessing:** Categorical data handling using **label or one-hot encoding**, null values handling.
- **Feature Selection:** Selecting feature and target variable(X,Y).
- **Data Splitting:** Splitting data into training and testing sets using **train_test_split**.
- **Model Selection:** Select regression models such as Multiple linear regression, decision tree, random forest etc...
- **Model Training:** Train the data with different models using **.fit()** method.
- **Model Evaluation:** Evaluate the model using metrics such as **R-squared Score**.
- **Make Predictions:** Use the best model to predict students final exam score.